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CIS 492/593
3 May. 2026

Introduction

The purpose of this project was to design, build, and test a functional drone using the Air:bit 2 platform, while also developing code that enables it to complete a specific task. The main objective was to program the drone to autonomously navigate to another team's rover and successfully capture an image.

The Air:bit 2 Drone is a programmable educational drone that can be controlled through a web-based platform using either block-based coding or JavaScript. By applying logical sequencing and precise timing, the drone can perform tasks such as movement, rotation, and image capture. Throughout this project, our team focused on developing a reliable flight path, maintaining stable movement, and accurately timing the camera function.

System Specifications

Overall Air:bit 2 Drone Specifications:

- Overall Length: about 120 mm (4.72 inches)
- Overall Width: about 110 mm (4.33 inches)
- Overall Height: about 30 mm (1.18 inches)
- Weight: about 90 grams (without camera module)
- Drone Type: programmable quadcopter drone
- Main Controller: BBC micro:bit

- Motors: 4 high-power drone motors
- Propellers: 4 active propellers (with replacements available)
- Battery: 3.7V rechargeable LiPo battery
- Flight Time: dependent on battery charge and added weight
- Remote Control: second micro:bit used as controller
- Frame Material: lightweight plywood frame
- Added Component: camera module (added by team)
- Speed: not officially specified

System Description:

The Air:bit 2 is a small programmable drone that uses two BBC micro:bits. One micro:bit is mounted on the drone and acts as the flight controller, while the second micro:bit is used as a remote controller. In this project, the drone was first tested under normal flight conditions and later modified by adding a camera module to expand its functionality and complete the mission objective.

Hardware Components:

- Air:bit 2 drone frame kit
- BBC micro:bit (x2)
- Air:bit 2 flight controller board
- 4 drone motors
- 8 propellers
- Rechargeable LiPo battery
- Propeller guards
- Screws, spacers, and mounting hardware
- Remote controller holder
- Camera module (added component)
- Replacement motor and propellers
- USB cable for programming and charging
- Battery charging connection



Software Specifications:

- Platform: Microsoft MakeCode
 - Programming Language: Block-based programming and JavaScript
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Testing and Results

The drone was tested multiple times throughout the development process to evaluate its performance and identify issues affecting functionality. Each test provided insight into specific problems, which were then addressed to improve overall operation.

During the initial testing phase, the primary issue was maintaining sufficient battery life. The drone required frequent charging, which limited testing time. To resolve this, a team member was assigned to ensure the drone was fully charged before each session, allowing for more consistent testing.

In early flight tests, one of the propellers failed to operate correctly. After inspection, it was discovered that the motor wires were loose. Once the wiring was secured, the propeller functioned properly, and the drone was able to generate balanced thrust.

Subsequent testing revealed that the drone struggled to achieve lift due to excess weight. The propeller guards were removed to reduce weight, which resulted in improved lift and successful takeoff.

During further testing, the drone exhibited instability and was difficult to control. The sensitivity of the remote control was reduced to improve handling. While this increased stability, it introduced a delay in throttle response. Adjusting the dealthrottle setting helped achieve better control and responsiveness during flight.

Additional testing identified issues with camera wiring. The wires were difficult to route after assembly, suggesting that the camera should have been installed earlier in the process.

In the final testing phase, the drone collided with a wall, resulting in broken propellers, a damaged motor, and signs of electrical damage. This highlighted the importance of controlled testing and safe operation.

Overall, testing led to significant improvements in stability, lift, and control, even though some limitations remained.

Challenges

Several challenges were encountered throughout the project. These included limited battery life, hardware issues such as loose wiring and damaged components, and instability during flight. Additionally, integrating the camera introduced complexity in both weight distribution and wiring.

These challenges required iterative problem-solving, testing, and adjustments. The experience demonstrated the importance of planning, careful assembly, and gradual testing when working with hardware systems.

Conclusion

This project successfully demonstrated the design, construction, and testing of a functional drone using the Air:bit 2 platform. The objective of autonomously navigating to a target and capturing an image required the integration of hardware components with embedded system control through programmed logic.

Throughout the project, the drone functioned as an embedded system, where software directly controlled physical components such as motors and the camera. This required careful attention to timing, responsiveness, and system reliability.

The challenges encountered highlighted the importance of safety considerations in embedded systems. Stable flight parameters, controlled throttle adjustments, and gradual testing were essential to prevent system failure and physical damage. These factors relate to safety constraints and system limitations that must be maintained during operation.

Despite these challenges, iterative improvements resulted in better performance and control. This project provided valuable insight into real-world embedded systems, where software and hardware must work together efficiently.

Future improvements could include implementing more advanced control algorithms, improving hardware durability, and adding safety checks to enhance reliability and performance.

Group Members

Hemanth Yarramsetti – Hardware & Assembly

Nhat Le – Flight Control & Calibration

Christian Deguzman – Flight Control

Het Patel – Hardware & Assembly

Prabesh Mishra – System Integration

Shreejil Patel – Testing

Victoria Tellez – Website design

Yousef Tantour – Programming & Report Writing

Sri Satya Phanindra Chitturi –Testing

Goutham Kondapani – Testing